Achieving Envy-Freeness with Limited Subsidies under Negative Dichotomous Valuations

Mainak*

Abstract

We study the problem of allocating indivisible *chores* among agents in a fair manner, with subsidies. Envy-free allocations of indivisible items are not guaranteed to exist, but envy-freeness can be achieved by additionally providing some subsidy to the agents; in the chores setting this subsidy is naturally read as compensation paid to the agents who absorb the workload. We study the subsidy required under *negative dichotomous* valuations, i.e., among agents whose marginal disutility for any chore is either zero or one: $v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\}$ for every agent i, every $S \subseteq M$ and every $g \in M \setminus S$. This is the exact mirror of the dichotomous goods model of Barman et al. [BKNS22], in which every marginal is 0 or 1 and every item is a good.

The mirror is not a relabelling. Subsidies are one-signed — money is always a good — so negating an instance negates the valuations but not the money, and the goods theory does not transfer by symmetry. We record what does survive: the Halpern–Shah characterisation of envy-freeable allocations [HS19] holds verbatim, since it assumes nothing about the sign or the structure of the valuations, and it delivers an *integral* minimum subsidy vector in the negative dichotomous model. We also record what the existing literature already settles: negative dichotomous valuations are doubly monotone under the two-sided normalisation, so the reduction of Kawase et al. [KMS+24] already yields a subsidy of at most $n - 1$ per agent and $n(n - 1)/2$ in total. The open question is therefore not whether a bounded subsidy exists, but whether the far stronger guarantee of [BKNS22] — a subsidy of exactly 0 or 1 to each agent, hence $n - 1$ in total — survives the passage from goods to chores. We show that the matching lower bound of $n - 1$ does survive.

[**TODO:** state the contribution once Approach 1 lands: either the $p \in \{0, 1\}^n$ analogue of [BKNS22, Theorem 4] together with a polynomial-time value-oracle algorithm, or the counterexample separating the two models. Then revise the two paragraphs above so that the abstract leads with the result rather than with the framing.]

1 Introduction

Discrete fair division studies the allocation of resources that cannot be fractionally assigned among agents with individual preferences, and sits at the interface of mathematical economics and computer science [BCE+16, End17]. The classical fairness criterion is *envy-freeness* [Fol67, Var74]: no agent should prefer the bundle assigned to another agent over her

*Affiliation and contact details to be filled in. [author block: confirm co-authors before circulating]

own. For indivisible items an envy-free allocation need not exist, as the standard one-item two-agent instance shows, and a large body of work therefore studies relaxations such as envy-freeness up to one good (EF1) [Bud11, LMMS04].

A second and complementary line of work retains *exact* envy-freeness and buys off the residual envy with money. Each agent i receives, in addition to her bundle, a subsidy $p_i \geq 0$ supplied by a third party, and one asks for an allocation together with a subsidy vector under which every agent weakly prefers her own bundle-plus-subsidy to that of every other agent. The question is how much money this requires. Maskin [Mas87] answered it for unit-demand agents with $m = n$: under the scaling in which no agent values any single item above 1, a total subsidy of $n - 1$ suffices. Halpern and Shah [HS19] moved to the multi-demand setting with arbitrary m, gave the characterisation of *envy-freeable* allocations that the whole line now runs on, showed that a subsidy of $(n-1)m$ is necessary and sufficient in the worst case when the allocation is handed to us, and conjectured that $n - 1$ suffices when we may choose the allocation. Brustle et al. [BDN+20] proved that conjecture in the stronger per-agent form — one dollar to each agent suffices for additive valuations, with the allocation simultaneously EF1 and balanced — and showed that for general monotone valuations $2(n-1)$ per agent suffices, so that the total subsidy is $O(n^2)$ and, remarkably, *independent of the number of items*. Barman et al. [BKNS22] then improved the monotone bound by a factor of n on the dichotomous subclass: when every marginal $v_i(S \cup \{g\}) - v_i(S)$ lies in $\{0, 1\}$, there is an allocation whose accompanying subsidy is exactly 0 or 1 for each agent, hence at most $n - 1$ in total, computable in polynomial time in the value-oracle model. Their bound assumes neither additivity nor submodularity nor even subadditivity, and it is tight: with n agents and a single unit-valued good, $n - 1$ agents must each be paid 1.

Chores. Every result above is about *goods*. This paper asks the mirrored question, in which every item is a *chore*: an object that an agent would rather not hold, so that acquiring it weakly decreases her utility. Chores are not an exotic variant. Shift rotations, on-call duty, teaching and refereeing load, committee service, maintenance tasks and the liabilities attached to an estate are all allocation problems in which the items are burdens rather than prizes, and in all of them money is already the standard instrument of repair — as overtime pay, hardship allowance, or a course-release budget. If anything the subsidy model is more natural for chores than for goods, since compensating somebody for taking on unwanted work needs less justification than paying somebody for the privilege of not receiving a prize.

Concretely, we study *negative dichotomous* valuations: for every agent i, every $S \subseteq M$ and every $g \in M \setminus S$,

$$v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\}.$$

Equivalently, each marginal $v_i(S \cup \{g\}) - v_i(S)$ is 0 or -1: every item is a chore for every agent, and taking on one more chore costs an agent either nothing or exactly one unit. This is the exact mirror of the model of [BKNS22], and we impose no further structure — in particular the valuations need not be additive, submodular, or subadditive.

Binary marginals are the natural model whenever a task either lands on an agent as a fresh unit of burden or is absorbed by a commitment she has already made. An engineer already staffing an on-call window absorbs a second incident in that window at no extra cost;

a referee already reading for a venue absorbs one more paper from the same batch; a driver already routed through a district absorbs an extra stop along the way. Whether a given chore is free depends in an arbitrary way on the rest of the agent's assignment, which is exactly why the model must tolerate non-additive and even non-subadditive valuations, and it is the counterpart of the scheduling example used by [BKNS22] to motivate dichotomous valuations for goods.

Why this is not a change of sign. It is tempting to expect that a chores result follows from the corresponding goods result by negating the valuations. It does not, for three reasons that we record here because they shape everything that follows.

First, money is one-signed. The subsidy vector is constrained to $p \in \mathbb{R}^n_+$, and negating an instance negates the valuations but leaves the subsidies where they are. The polyhedron of feasible subsidy vectors is therefore a genuinely different object in the two models, and no formal duality carries a bound across.

Second, the obvious transformation is available only for two agents. Writing $c_i := -v_i$ for the cost form of a negative dichotomous valuation, the complement $\tilde{v}_i(S) := c_i(M) - c_i(M \setminus S)$ is again dichotomous in the sense of [BKNS22], so the class is closed under complementation. But the complement transformation respects the allocation only when $M \setminus A_j = A_i$, that is, only when $n = 2$; see Remark 6. For $n \geq 3$ there is no such reduction, and that is where the content of the problem lies.

Third, and most tellingly, envy flows the other way. Under goods, envy concentrates *into* the agent holding the valuable items: many agents envy one, and the tight instance of [BKNS22] pays $n-1$ agents one dollar each so that they stop envying the single agent who received the good. Under chores, envy emanates *out of* the agent carrying the load: one agent envies many. Since the minimum subsidy to an agent is the weight of the heaviest directed path leaving her in the envy graph [HS19, Theorem 2], the two situations are not interchangeable, and one should not assume in advance that the same bound is the right target.

What is already known, and what is open. Two facts delimit the problem before any work is done, and we state them explicitly so that the contribution can be measured against them.

On the one hand, existence of a bounded subsidy is not in question. Negative dichotomous valuations are doubly monotone — every item is a chore for every agent, which is a special case of every item being a good or a chore for each agent — and they satisfy the two-sided normalisation $|v_i(S \cup \{e\}) - v_i(S)| \leq 1$. The reduction of Kawase et al. [KMS[+]24], which converts any EF1 allocation into an envy-free one, therefore applies off the shelf and yields a subsidy of at most $n-1$ per agent and $n(n-1)/2$ in total. [**VERIFY:** this containment and the resulting bound against Reading_9 directly; confirm in particular which EF1 convention [KMS[+]24] requires, since the goods-only and two-sided forms of EF1 differ once chores are present]

On the other hand, the lower bound of the goods model survives intact.

Example 1 (The $n-1$ lower bound transfers)**.** Take n agents and $m = n-1$ chores, with $c_i(S) = |S|$ for every agent i — identical, binary, additive costs, which are in particular negative dichotomous. Every complete allocation leaves at least one agent empty-handed. Under the balanced allocation, in which $n-1$ agents hold one chore each and one agent holds nothing, each of the $n-1$ loaded agents envies the empty agent by exactly 1 and no other envy is present, so the minimum subsidy assigns 1 to each loaded agent and 0 to the empty one, for a total of $n-1$. No allocation does better: any complete allocation gives some agent a bundle of size $k \geq 1$ while some other agent is empty, and that agent's heaviest outgoing path already has weight k.

So the target statement is precisely the analogue of [BKNS22, Theorem 4]: that under negative dichotomous valuations there is an allocation admitting a subsidy vector $p \in \{0,1\}^n$, hence a total subsidy of at most $n-1$, computable in polynomial time with value-oracle access. Example 1 shows this would be tight. Between the $n-1$ per agent that [KMS+24] already gives and the $\{0,1\}$ per agent that we are asking for lies a factor of n in the total, exactly the gap that [BKNS22] closed on the goods side. Our objective is to close it on the chores side or to exhibit an instance showing that it cannot be closed.

Our Results. [**TODO:** write this paragraph once Approach 1 is settled. It should mirror the "Our Results" paragraph of [BKNS22]: one sentence stating the theorem, one sentence on the algorithmic guarantee in the value-oracle model, one sentence on tightness pointing at Example 1, and one sentence saying which structural assumptions are *not* needed. If the outcome is a separation instead, state the counterexample and the largest subsidy bound that does survive.]

Related Work. The subsidy line for goods is described above. Within it, the dichotomous subclass has attracted separate attention: Halpern and Shah [HS19] settle binary additive valuations with a total subsidy of $n-1$, Goko et al. [GIK+21] obtain the analogous bound for binary submodular valuations together with a strategy-proof mechanism, and Barman et al. [BKNS22] subsume both by dropping every structural assumption beyond binary marginals. Dichotomous preferences are themselves a well-studied restriction in social choice [BMS05, BEF21]. On the computational side, Caragiannis and Ioannidis [CI21] show that minimising the subsidy is NP-hard and give a fully polynomial-time approximation scheme for a constant number of agents, and Narayan et al. [NSV21] study the welfare cost of achieving envy-freeness with transfers.

The paper closest to the present setting is that of Kawase et al. [KMS+24], which is the only work in this line that admits chores at all. Working with doubly monotone valuations under the two-sided normalisation, it decouples the two jobs that [BDN+20] performed together: rather than constructing a specific allocation and showing that it is cheap to subsidise, it takes *any* EF1 allocation as input and bounds the subsidy needed to convert it, obtaining $n-1$ per agent and $n(n-1)/2$ in total. Since EF1 allocations are known to exist for doubly monotone valuations, the wider class comes along at no cost. That reduction is what makes the present question well-posed rather than open-ended, and it is the baseline we are trying to beat.

The complementary repair — no money, weaker fairness notion — is pursued for binary valuations by Bu et al. [BSY23], who show that EFX allocations always exist and can be computed in polynomial time for general binary valuations, thus extending the matroid-rank result of Babaioff et al. [BEF21]. Under binary valuations one therefore has a clean dichotomy on the goods side: either pay 0 or 1 per agent and obtain exact envy-freeness [BKNS22], or pay nothing and obtain EFX [BSY23]. Whether either branch survives the passage to chores is open; this paper addresses the first.

[**TODO:** a literature sweep of the chores-side fair division literature is still outstanding. Nothing in the corpus $R1$–$R9$ covers indivisible chores except [KMS+24], and the related-work paragraph above therefore says nothing about EF1/EFX for chores, about the goods-and-chores literature, or about prior uses of subsidies for task assignment. Add those readings to `References/`, extend `paper_map_R1_to_R9.md`, and rewrite this paragraph before circulating.]

2 Notation and Preliminaries

We study the allocation of m indivisible items among n agents, with subsidies. Write $N = [n]$ for the set of agents and M for the set of items, $|M| = m$. Each agent $i \in N$ has a valuation $v_i : 2^M \to \mathbb{R}$ with $v_i(\emptyset) = 0$, and we represent an instance by the tuple $\langle N, M, \{v_i\}_{i\in N}\rangle$. Algorithms operate in the standard *value-oracle* model: for each v_i we assume only an oracle that returns $v_i(S)$ for a queried set $S \subseteq M$, and running time is measured in n, m and the number of oracle calls.

An *allocation* $\mathcal{A} = (A_1, \ldots, A_n)$ is an ordered tuple of pairwise disjoint *bundles*, with A_i assigned to agent i. The allocation is *complete* if $\bigcup_{i\in N} A_i = M$ and *partial* otherwise. The ordering matters throughout: much of the theory below concerns permuting a fixed family of bundles among the agents, and for a permutation $\sigma \in \mathbb{S}_n$ we write $\mathcal{A}_\sigma := (A_{\sigma(1)}, \ldots, A_{\sigma(n)})$ for the allocation in which agent i receives $A_{\sigma(i)}$. The utilitarian welfare of $\mathcal{A}$ is $\mathrm{SW}(\mathcal{A}) = \sum_{i\in N} v_i(A_i)$.

Definition 2 (Envy-Freeness)**.** *An allocation $\mathcal{A} = (A_1, \ldots, A_n)$ is* envy-free *(EF) iff $v_i(A_i) \geq v_i(A_j)$ for all agents $i, j \in N$.*

Envy-free allocations of indivisible items need not exist, so we allow a third party to supplement the allocation with money. Agents have quasi-linear utilities: agent i receiving bundle A_i and subsidy p_i enjoys utility $v_i(A_i) + p_i$, with money entering linearly and at the same exchange rate for every agent.

Definition 3 (Envy-Free Solution)**.** *An allocation $\mathcal{A} = (A_1, \ldots, A_n)$ and a subsidy vector $p = (p_1, \ldots, p_n) \in \mathbb{R}^n_+$ constitute an* envy-free solution *$(\mathcal{A}, p)$ iff $v_i(A_i) + p_i \geq v_i(A_j) + p_j$ for all $i, j \in N$.*

An allocation $\mathcal{A}$ is *envy-freeable* if some $p \in \mathbb{R}^n_+$ makes $(\mathcal{A}, p)$ an envy-free solution; this is a property of the allocation alone. Following [BKNS22] we write $\mathrm{M}(p) := \{i \in N \mid p_i \geq p_j \text{ for all } j \in N\}$ for the set of maximally subsidised agents.

We will occasionally need the standard relaxation of envy-freeness. Note that two inequivalent forms of it appear in this literature: the goods-only form of [HS19], which removes one item from the envied bundle, and the two-sided form used by [KMS+24], which removes one item from *either* bundle. They coincide when all items are goods and differ once chores are present, and only the two-sided form supports the doubly monotone results. We adopt the two-sided form.

Definition 4 (EF1, two-sided form)**.** *An allocation $\mathcal{A}$ is* envy-free up to one item *(EF1) iff for all $i, j \in N$ there exists $X \subseteq A_i \cup A_j$ with $|X| \leq 1$ such that $v_i(A_i \setminus X) \geq v_i(A_j \setminus X)$.*

2.1 Negative Dichotomous Valuations

Definition 5 (Negative Dichotomous Valuation)**.** *A valuation $v_i : 2^M \to \mathbb{R}$ with $v_i(\emptyset) = 0$ is* negative dichotomous *iff*

$$v_i(S) - v_i(S \cup \{g\}) \ \in \ \{0, 1\} \qquad \text{for every } S \subseteq M \text{ and every } g \in M \setminus S,$$

equivalently iff every marginal $\Delta_i^+(S, g) := v_i(S \cup \{g\}) - v_i(S)$ lies in $\{-1, 0\}$.

Every item is thus a chore for every agent: v_i is non-increasing, so $v_i(S) \geq v_i(T)$ whenever $S \subseteq T$. We stress that nothing further is assumed — the valuations need not be additive, submodular, or subadditive — which is the generality at which [BKNS22] works on the goods side.

It is often more convenient to work with the non-negative *cost form*

$$c_i \ := \ -v_i, \qquad \text{so} \qquad c_i(\emptyset) = 0, \quad c_i(S) \leq c_i(T) \text{ for } S \subseteq T, \quad c_i(S \cup \{g\}) - c_i(S) \in \{0, 1\}\,.$$

The correspondence $v_i \leftrightarrow c_i$ is a bijection between negative dichotomous valuations and dichotomous valuations in the sense of [BKNS22]: the class we study is exactly the pointwise negation of the class they study. In cost form an envy-free solution is the requirement $c_i(A_i) - p_i \leq c_i(A_j) - p_j$ for all i, j, read as "my own burden net of my compensation is no worse than what I perceive of yours". We use whichever of v_i and c_i is clearer locally and always state which.

The negation, however, does not carry results across, for the reasons given in Section 1. The following remark isolates the one case in which a transformation does exist, and shows why it is uninformative.

Remark 6 (Complementation, and why it stops at $n = 2$)**.** For a negative dichotomous v_i with cost form c_i, define the *complement* $\tilde{v}_i(S) := c_i(M) - c_i(M \setminus S)$. Then $\tilde{v}_i(\emptyset) = 0$ and, for $g \notin S$, writing $T := M \setminus (S \cup \{g\})$,

$$\tilde{v}_i(S \cup \{g\}) - \tilde{v}_i(S) \ = \ c_i(M \setminus S) - c_i(T) \ = \ c_i(T \cup \{g\}) - c_i(T) \ \in \ \{0, 1\}\,,$$

so $\tilde{v}_i$ is dichotomous in the sense of [BKNS22]. The class is therefore closed under complementation, without any submodularity assumption. Now let $n = 2$ and let $\mathcal{A} = (A_1, A_2)$ be complete, so that $M \setminus A_2 = A_1$ and $M \setminus A_1 = A_2$. Then $-c_i(A_i) = \tilde{v}_i(A_j) - c_i(M)$ and $-c_i(A_j) = \tilde{v}_i(A_i) - c_i(M)$, so $(\mathcal{A}, p)$ is an envy-free solution for the chores instance $\{v_i\}$ if and only if $((A_2, A_1), p)$ is an envy-free solution for the goods instance $\{\tilde{v}_i\}$: swap the bundles and leave each subsidy attached to its agent. Two-agent chore division is thus exactly two-agent goods division, and [BKNS22] settles it. The step $M \setminus A_j = A_i$ has no analogue for $n \geq 3$, where the complement of one bundle is a union of several others, and no reduction of this kind is available.

2.2 The Envy Graph and the Halpern–Shah Characterisation

For an allocation $\mathcal{A}$, the *envy graph* $G_{\mathcal{A}}$ is the complete weighted directed graph on vertex set N in which the arc (i, j) carries weight

$$w_{\mathcal{A}}(i, j) \ := \ v_i(A_j) - v_i(A_i) \ = \ c_i(A_i) - c_i(A_j),$$

the envy that agent i has towards agent j. The weight of a directed path P is $w_{\mathcal{A}}(P) = \sum_{(i,j)\in P} w_{\mathcal{A}}(i,j)$, and $\ell_{\mathcal{A}}(i)$ denotes the maximum weight of any path in $G_{\mathcal{A}}$ starting at i, the empty path of weight 0 included.

The two results below are the engine of the entire subsidy line. Both are stated by Halpern and Shah [HS19] for additive valuations, but their proofs use only the arc weights and permutations of the bundles — neither additivity nor monotonicity nor any sign condition on v_i enters — and [BKNS22] invokes them in exactly this generality for (non-additive) dichotomous valuations. They therefore apply verbatim to negative dichotomous valuations, and we use them without further comment.

Theorem 7 ([HS19])**.** *For any allocation $\mathcal{A} = (A_1, \ldots, A_n)$, the following are equivalent.*

(i) $\mathcal{A}$ is envy-freeable.

(ii) $\mathcal{A}$ maximises utilitarian welfare across all reassignments of its own bundles among the agents: for every permutation σ over N, $\sum_{i\in N} v_i(A_i) \geq \sum_{i\in N} v_i(A_{\sigma(i)})$.

(iii) The envy graph $G_{\mathcal{A}}$ has no positive-weight directed cycle.

Theorem 8 ([HS19])**.** *For any envy-freeable allocation $\mathcal{A}$, the subsidy vector p^* given by $p^*_i := \ell_{\mathcal{A}}(i)$ realises an envy-free solution $(\mathcal{A}, p^*)$, and $p^*_i \leq p_i$ for every $i \in N$ and every p such that $(\mathcal{A}, p)$ is an envy-free solution. It can be computed in strongly polynomial time by running an all-pairs longest-path computation on $G_{\mathcal{A}}$.*

Condition (ii) of Theorem 7 has the standing consequence that an envy-freeable allocation can always be manufactured from an arbitrary one: fix the bundles and reassign them according to a maximum-weight perfect matching between agents and bundles. What is at issue is never the existence of an envy-freeable allocation, only the size of the subsidy that Theorem 8 then charges for it.

Specialising to our model gives the following, which we use throughout.

Observation 9 (Integrality)**.** *Let $\{v_i\}_{i\in N}$ be negative dichotomous. Then $v_i(S) \in \mathbb{Z}_{\leq 0}$ for every i and every $S \subseteq M$; every arc weight $w_{\mathcal{A}}(i,j)$ is an integer; and for every envy-freeable allocation $\mathcal{A}$ the minimum subsidy vector satisfies $p^* \in \mathbb{Z}^n_{\geq 0}$.*

Proof Fix i and $S = \{g_1, \ldots, g_k\}$. Telescoping from $v_i(\emptyset) = 0$ along any ordering of S writes $v_i(S)$ as a sum of k marginals, each of which lies in $\{-1, 0\}$ by Definition 5; hence $v_i(S) \in \mathbb{Z}$ and $-k \leq v_i(S) \leq 0$. Consequently each arc weight $w_{\mathcal{A}}(i,j) = v_i(A_j) - v_i(A_i)$ is a difference of integers, and the weight of any directed path, being a finite sum of arc weights, is an integer. By Theorem 8, $p^*_i = \ell_{\mathcal{A}}(i)$ is the maximum such weight over paths starting at i, and it is non-negative because the empty path has weight 0. □

Observation 9 is what makes the target statement — a subsidy vector in $\{0,1\}^n$ — a statement about a *bound* rather than about rounding: once p^* is known to be integral, showing $p^*_i \leq 1$ for every i is the same as showing $p^* \in \{0,1\}^n$. The same observation underlies the corresponding step in [BKNS22], where the arc weights of a dichotomous instance are likewise integers.

3 Our Results

[**TODO:** empty by design — fill in once Approach 1 has produced a theorem. The statement being aimed at is the negative dichotomous analogue of [BKNS22, Theorem 4]: an envy-free solution $(\mathcal{A}, p)$ with $p \in \{0,1\}^n$, hence total subsidy at most $n-1$, computable in polynomial time given value-oracle access. Example 1 already supplies the matching lower bound.]

4 Approach 1

[**TODO:** awaiting content.]

5 Conclusion and Future Work

[**TODO:** write last. Directions that are already visible, independent of how Approach 1 turns out: (i) the mixed model, in which some items are goods and others chores for the same agent, with all marginals in $\{-1,0,1\}$ — the dichotomous restriction of the doubly monotone setting of [KMS+24]; (ii) whether an EF1 guarantee can be obtained simultaneously, which [BKNS22, Appendix C] shows their goods algorithm does *not* provide; (iii) the weighted version, where [HS19]'s characterisation fails outright and the envy graph must be rebuilt.]

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